The Dutch grid in our model: maps, sources and assumptions

What the grid we use looks like today and in 2035, where every element comes from (JAO, Juan, Amir/TenneT, our own assumptions), how ratings are set, and how it compares with TenneT's plans. 2026-10-26. Grids: `jao_core_v29-vvltr2` (2025 validation) and `jao_core_v29-vvltr2-2035-tennet-tlb3-rt4k` (2035 dispatch, used by the RenewaFLEX session).

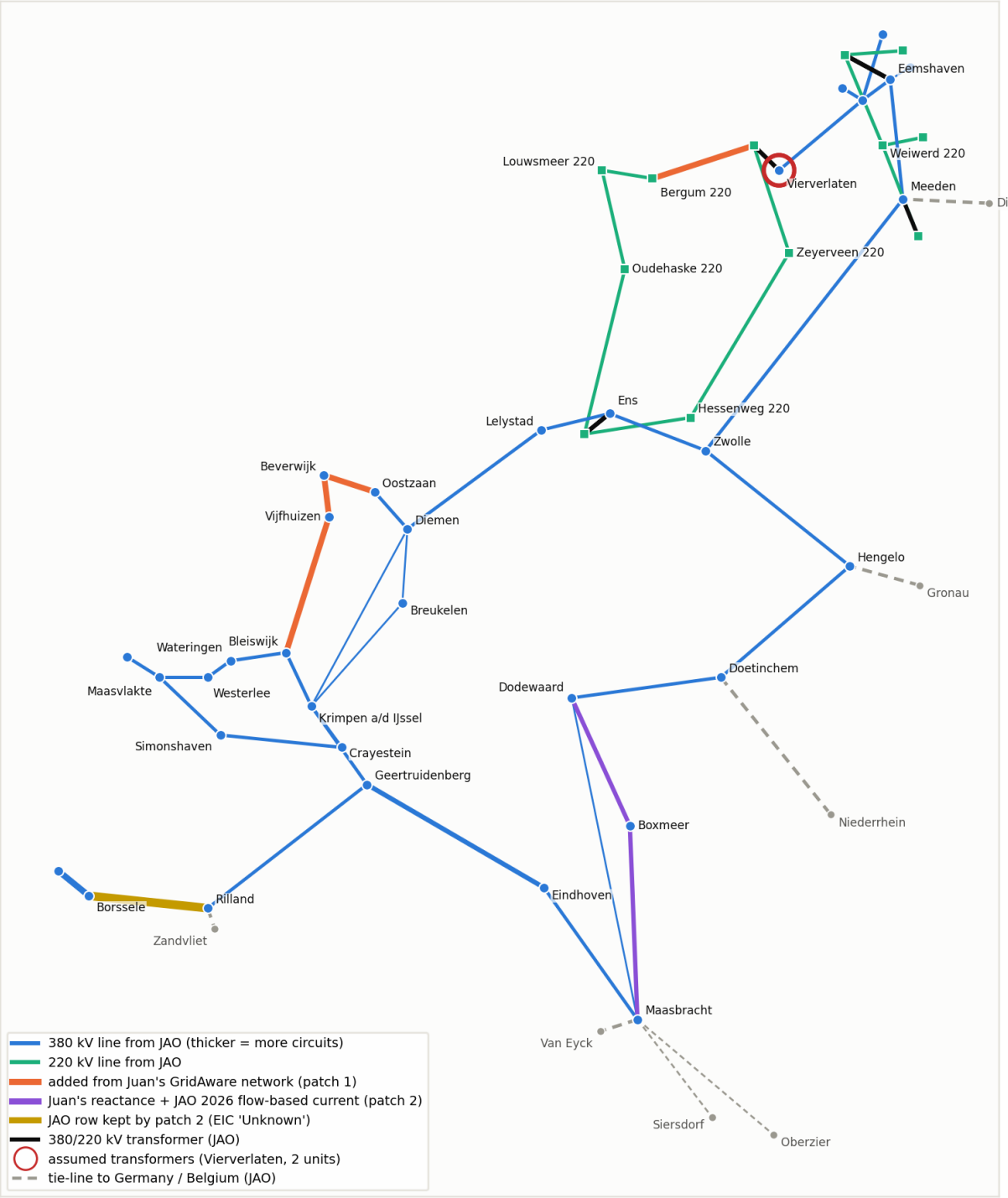
1. In short

- **Base:** JAO's Core Static Grid Model (March 2026). Of the 116 Dutch rows in today's grid (lines, tie-lines and transformers touching the Netherlands), 103 come straight from JAO; 8 were added from Juan's GridAware network, 2 combine Juan's reactance with JAO's 2026 flow-based current, 1 is a JAO row our first build dropped, and 2 are assumed transformers (Vierverlaten).
- **Today's grid:** 33 Dutch 380 kV and 12 220 kV buses, 72 380 kV and 22 220 kV circuits, 10 transformers, 12 tie-line circuits.
- **2035 grid:** today's grid plus the IP2026 projects in realisation (TenneT's own rule), the Tilburg loop-in and the 4 kA fix for two finished upgrades: 23 rows changed or added, 3 removed.
- **Ratings:** MW = sqrt(3) x 400 kV (or 225 kV) x current, the convention of JAO's own flow-based data; JAO's current = mean of its listed seasonal values. In 2035, 16 of 36 Dutch 380 kV corridors are still below 4,000 A per circuit (map 3).
- **Against TenneT:** our 2035 grid follows TenneT's own planning rule (projects in realisation). TenneT's IP26 layer has 7 380/220 kV projects due by 2035 that are not yet in realisation and therefore not in our grid, and 14 due after 2035 (map 2).

2. Where the data comes from

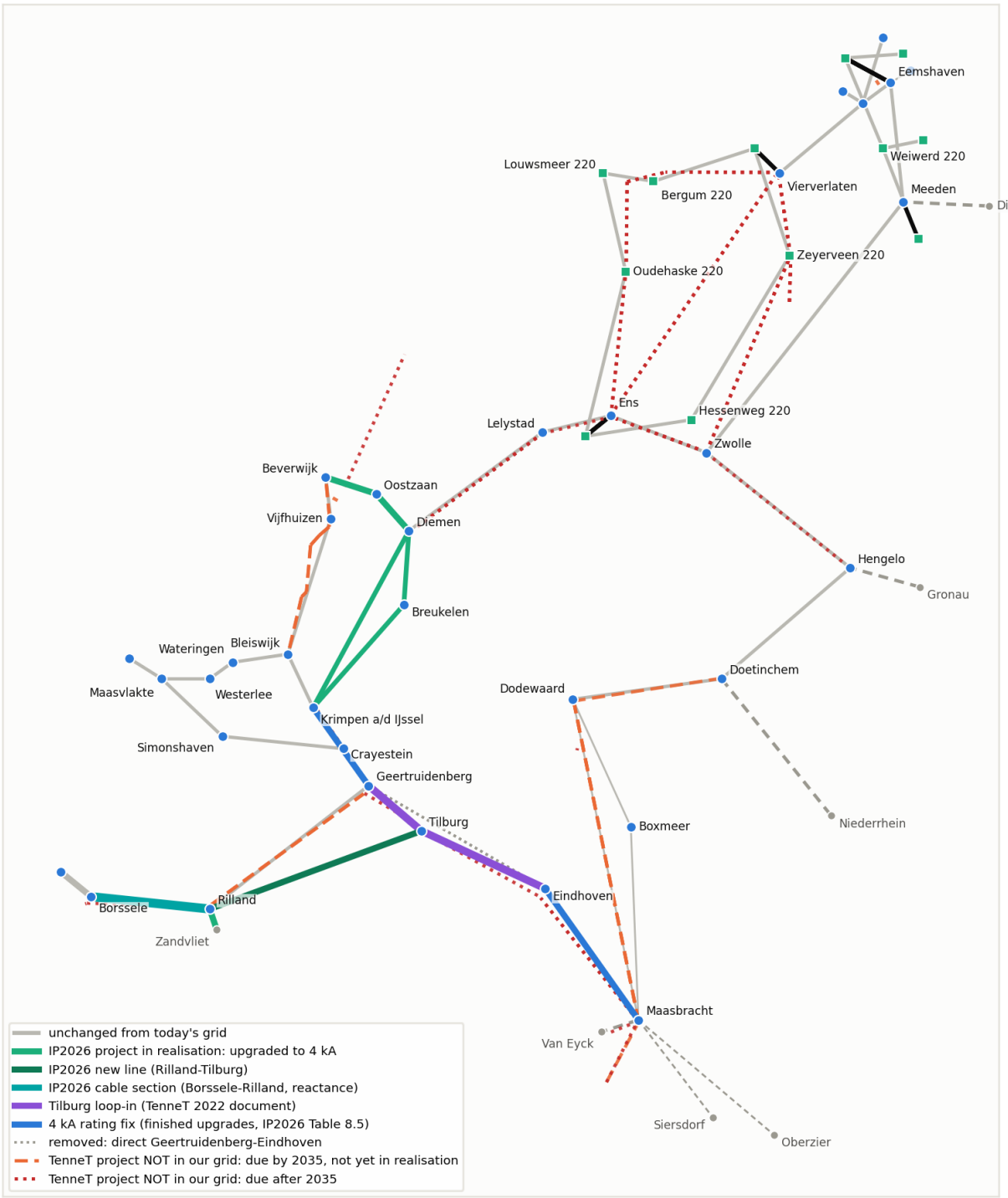
Source	What we took	How it enters	Terms / note
JAO Core Static Grid Model, 9th release, March 2026	All lines, tie-lines, transformers, currents, reactances, lengths, substation names; coordinates from JAO's publication page	Builder (archived script 10) -> workbook; fixes v23-v27 (none Dutch); file in inputs/raw/	JAO publication data; check terms before redistributing
JAO 2026 flow-based data (publication tool)	Evidence: which Dutch elements JAO monitors; the current of Maasbracht-Boxmeer-Dodewaard (2,498 A, July 2026)	Patch jao2026_nl_2	Analysis only
Juan: GridAware full network (full_network.csv)	Missing Dutch elements: Randstad ring Beverwijk-Vijfhuizen-Bleiswijk, Beverwijk-Oostzaan, Bergum-Vierverlaten 220 kV; reactance and length of Maasbracht-Boxmeer-Dodewaard. Only current, reactance and length taken.	Patches jao2026_nl (v28) and jao2026_nl_2 (v29); docs/14	Internal project data (copy in NewLines_IP26 update/DATA)
Amir: IP26 network package (NewLines_IP26)	Mapping of IP2026 line projects to grid edges; his inclusion rule (realisation + preparation); 'reinforced rating' 4 kA; later update with cables and partial upgrades	Decision table ip2026_nl_2035_decisions (column amir_edge); docs/17-19	Built on Juan's network and TenneT's IP26 layer
TenneT Investeringsplan 2026 Net op land + annexes	Project phases and dates, the 4 kA statement (s7.4), Table 8.5 finished upgrades, bottleneck annexes	Decision table; 4 kA patch; G6/G8/G9 (RenewaFLEX)	Public PDF (IP2026/)
TenneT IP26 ArcGIS layers (Amir's DATA folder)	Project lines and stations with phase and commissioning window	Map 2; decision table	TenneT, all rights reserved: local use only
TenneT 2022 notice 'hoogspanningsstation Tilburg'	Loop-in of the existing 380 kV line into Tilburg, 3 circuits by 2035	Patch ip2026_nl_2035_tilburg_loopin_3c	Public document (copy in the patch folder)
Our own assumptions	Vierverlaten 380/220 kV transformers (no data anywhere); X of new and split lines	Patches nl_vierverlaten_tr_assumed_2u, Tilburg, Rilland-Tilburg	Labelled ASSUMED, section 5

3. Map 1 — today's grid, by source



Positions from the workbook's substation coordinates (about 0.01 degree); stations at the same place drawn slightly apart; no coastline. Coloured lines: corridors with at least one circuit that does not come straight from JAO (e.g. Borssele-Rilland: circuit P only; Maasbracht-Boxmeer-Dodewaard: circuit Z only).

4. Map 2 — the 2035 grid: what changed and what TenneT plans beyond it



TenneT projects drawn as straight lines between the ends of each project path in TenneT's IP26 layer (schematic routes). Categories as in the RenewaFLEX comparison (nl_merge G6). 'In realisation' is TenneT's phase label in IP2026.

5. Elements that do not come straight from JAO

5.1 Today's grid (patches on top of JAO)

Element	Source	Current A	MW	X ohm	Basis
TR Vierverlaten 380kV TR 426-assumed	Assumed	1,140	790	31.53	ASSUMED: median Dutch JAO 380/220 kV unit
TR Vierverlaten 380kV TR second-assumed	Assumed	1,140	790	31.53	ASSUMED: median Dutch JAO 380/220 kV unit
Borssele-Rilland380 P	JAO raw (patch 2)	4,000	2,771	10.02	JAO raw row (4,000 A both periods), EIC 'Unknown' so the first build dropped it
Bergum-Vierverlaten220 W	Juan	2,500	974	8.54	Juan's current; X and length from Juan
Bergum-Vierverlaten220 Z	Juan	2,500	974	8.54	Juan's current; X and length from Juan
Beverwijk-Oostzaan380 W	Juan	3,300	2,286	4.35	Juan's current; X and length from Juan
Beverwijk-Oostzaan380 Z	Juan	3,300	2,286	4.35	Juan's current; X and length from Juan
Beverwijk-Vijfhuizen380 W	Juan	3,000	2,078	2.86	Juan's current; X and length from Juan
Beverwijk-Vijfhuizen380 Z	Juan	3,000	2,078	2.86	Juan's current; X and length from Juan
Bleiswijk-Vijfhuizen380 O	Juan	3,000	2,078	9.92	Juan's current; X and length from Juan
Bleiswijk-Vijfhuizen380 P	Juan	3,000	2,078	9.92	Juan's current; X and length from Juan
Boxmeer-Dodewaard380 Z	Juan+JAO-FB	2,498	1,731	15.86	current 2,498 A from JAO 2026 flow-based (July); X and length from Juan
Maasbracht-Boxmeer380 Z	Juan+JAO-FB	2,498	1,731	15.27	current 2,498 A from JAO 2026 flow-based (July); X and length from Juan

5.2 The 2035 changes

Element	Change	A today	A 2035	X ohm	Basis
Krimpen aan den IJssel-Geertruidenberg380 W	4 kA (IP2026 Table 8.5)	3,300	4,000	9.86	finished 'Beter Benutten' upgrade, 2 x 2,635 MVA (4 kA); JAO 2026 flow-based data already uses 4 kA (Krimpen-Geertruidenberg)
Krimpen aan den IJssel-Geertruidenberg380 Z	4 kA (IP2026 Table 8.5)	3,300	4,000	9.86	finished 'Beter Benutten' upgrade, 2 x 2,635 MVA (4 kA); JAO 2026 flow-based data already uses 4 kA (Krimpen-Geertruidenberg)
Maasbracht-Eindhoven380 W	4 kA (IP2026 Table 8.5)	3,300	4,000	13.40	finished 'Beter Benutten' upgrade, 2 x 2,635 MVA (4 kA); JAO 2026 flow-based data already uses 4 kA (Krimpen-Geertruidenberg)
Maasbracht-Eindhoven380 Z	4 kA (IP2026 Table 8.5)	3,300	4,000	13.40	finished 'Beter Benutten' upgrade, 2 x 2,635 MVA (4 kA); JAO 2026 flow-based data already uses 4 kA (Krimpen-Geertruidenberg)
Borssele-Rilland380 G	IP2026 cable section (X)	3,238	3,238	7.28	IP2026 A-1002896 cable section; X reduced, rating unchanged (needs confirmation)
Borssele-Rilland380 Z	IP2026 cable section (X)	3,238	3,238	7.28	IP2026 A-1002896 cable section; X reduced, rating unchanged (needs confirmation)
Rilland-Tilburg380 W	IP2026 new line	—	4,000	21.29	IP2026 A-1002678; 4,000 A; X = 78.1 km x 0.273 ohm/km (median Dutch JAO 380 kV, ASSUMED)
Rilland-Tilburg380 Z	IP2026 new line	—	4,000	21.29	IP2026 A-1002678; 4,000 A; X = 78.1 km x 0.273 ohm/km (median Dutch JAO 380 kV, ASSUMED)
Beverwijk-Oostzaan380 W	IP2026 upgrade to 4 kA	3,300	4,000	4.35	IP2026 project in realisation; 4,000 A (IP2026 s7.4, Amir's reinforced rating)
Beverwijk-Oostzaan380 Z	IP2026 upgrade to 4 kA	3,300	4,000	4.35	IP2026 project in realisation; 4,000 A (IP2026 s7.4, Amir's reinforced rating)
Breukelen Kortrijk-Diemen380 W	IP2026 upgrade to 4 kA	2,910	4,000	5.43	IP2026 project in realisation; 4,000 A (IP2026 s7.4, Amir's reinforced rating)
Krimpen aan den IJssel-Breukelen Kortrijk380 W	IP2026 upgrade to 4 kA	2,884	4,000	10.25	IP2026 project in realisation; 4,000 A (IP2026 s7.4, Amir's reinforced rating)
Krimpen aan den IJssel-Diemen380 Z	IP2026 upgrade to 4 kA	2,884	4,000	15.70	IP2026 project in realisation; 4,000 A (IP2026 s7.4, Amir's reinforced rating)
Oostzaan-Diemen380 G	IP2026 upgrade to 4 kA	2,944	4,000	4.21	IP2026 project in realisation; 4,000 A (IP2026 s7.4, Amir's reinforced rating)

Element	Change	A today	A 2035	X ohm	Basis
Oostzaan-Diemen380 Z	IP2026 upgrade to 4 kA	2,944	4,000	4.21	IP2026 project in realisation; 4,000 A (IP2026 s7.4, Amir's reinforced rating)
RILLAND – ZANDVLIET GREY/29	IP2026 upgrade to 4 kA	3,790	4,000	2.05	IP2026 project in realisation; 4,000 A (IP2026 s7.4, Amir's reinforced rating)
RILLAND – ZANDVLIET WHITE/30	IP2026 upgrade to 4 kA	3,790	4,000	2.05	IP2026 project in realisation; 4,000 A (IP2026 s7.4, Amir's reinforced rating)
Geertruidenberg-Tilburg380 G	Tilburg loop-in	—	2,884	6.41	TenneT 2022; X and length split 0.337/0.663 by straight-line distance (ASSUMED); rating of the old line
Geertruidenberg-Tilburg380 W	Tilburg loop-in	—	2,884	6.41	TenneT 2022; X and length split 0.337/0.663 by straight-line distance (ASSUMED); rating of the old line
Geertruidenberg-Tilburg380 Z	Tilburg loop-in	—	2,884	6.45	TenneT 2022; X and length split 0.337/0.663 by straight-line distance (ASSUMED); rating of the old line
Tilburg-Eindhoven380 G	Tilburg loop-in	—	2,884	12.62	TenneT 2022; X and length split 0.337/0.663 by straight-line distance (ASSUMED); rating of the old line
Tilburg-Eindhoven380 W	Tilburg loop-in	—	2,884	12.62	TenneT 2022; X and length split 0.337/0.663 by straight-line distance (ASSUMED); rating of the old line
Tilburg-Eindhoven380 Z	Tilburg loop-in	—	2,884	12.68	TenneT 2022; X and length split 0.337/0.663 by straight-line distance (ASSUMED); rating of the old line
Geertruidenberg-Eindhoven380 G	removed (loop-in)	2,884	—	—	replaced by Geertruidenberg-Tilburg and Tilburg-Eindhoven
Geertruidenberg-Eindhoven380 W	removed (loop-in)	2,884	—	—	replaced by Geertruidenberg-Tilburg and Tilburg-Eindhoven
Geertruidenberg-Eindhoven380 Z	removed (loop-in)	2,884	—	—	replaced by Geertruidenberg-Tilburg and Tilburg-Eindhoven

6. Assumptions, in one place

Assumption	Value	Why	Effect if wrong
Rating voltage	MW = sqrt(3) x 400 kV (225 kV) x current	JAO's own convention (Fmax / sqrt(3) Imax = 400 kV in its 2026 flow-based data)	None for comparisons with JAO; about 5 % above nominal-voltage MW
Current of a JAO line	Mean of the listed seasonal currents	One rating per line in the model (seasonal ratings tested, not adopted)	Summer limits about 10-15 % lower than used
Vierverlaten transformers	2 units, X 31.529 ohm, 1,140 A, 790 MW each	The coupling exists (IP2026 bottleneck 1578, asset VVL TR426) but no values in any file	Changes how the northern 220 kV ring shares flow (26-31 % of Vierverlaten-Ens)
New / split line reactance	0.273 ohm/km (median Dutch JAO 380 kV); Tilburg split by straight-line distance	No route data for new lines	Small: reactance shares of parallel paths
Maasbracht-Boxmeer-Doddewaard current	2,498 A (JAO July 2026)	JAO's January value 2,875 A, Juan 2,000/2,875 A also exist	This corridor's limit
Krimpen-Geertruidenberg, Maasbracht-Eindhoven 2035	4,000 A	IP2026 Table 8.5 finished upgrades; end-station works before 2035	Main 2035 bottleneck limit (3,010 -> 3,496 MW)
Selection rule 2035	TenneT's rule: realisation-phase projects only	TenneT's own planning rule (IP2026 s5.3); Amir's rule also takes preparation-phase projects	Leaves out projects that may be in service by 2035 (section 7)
Coordinates	JAO publication / IP26 points, rounded to 0.01 degree	Drawing only	None on results

7. What came from Juan, what came from Amir

	Juan (GridAware)	Amir (IP26 network package)
What it is	A full Dutch network file (full_network.csv): stations, lines, R, X, B, ratings	Scripts that turn TenneT's IP2026 line projects into grid edges on top of Juan's network, per scenario and horizon
Used in our grid	Yes: 8 rows of patch 1 (Randstad ring, Beverwijk-Oostzaan, Bergum-Vierverlaten 220) and the reactance/length of the Maasbracht-Boxmeer-Dodewaard circuit (patch 2)	Yes, indirectly: his project-to-edge mapping and 4 kA 'reinforced rating' fed the 2035 decision table; we applied TenneT's rule, not his
Not used	Juan's R and B (kept in the patch metadata); Juan's earlier TNO MATLAB network (no transformers, stations merged) — we keep JAO's transformers	His inclusion rule (realisation + preparation): built as a variant (v29-2035-amir), not used; his later update (cables, partial upgrades, Juan's median cable rating 1,481 A): reviewed, not adopted — it would halve Borssele-Rilland and treat 88-99.9 % traces as 'partial'
Where documented	docs/14; patch CHANGELOGs	docs/17, 18, 19; decision_table.csv; provenance (review of the update)

Amir's rule changes these lines in the 2035 grid (they are not in ours): Dodewaard-Doetinchem, Maasbracht-Boxmeer-Dodewaard and Geertruidenberg-Rilland upgraded to 4 kA, and a third Eemshaven Oudeschip-Eemshaven circuit (decision table rows D2035-009 to -013 and -024 to -026). It also counts the stations Einighausen and A9-Zuid, but those change no flows in a 380/220 kV model and are skipped under both rules (-008, -016).

8. Comparison with TenneT

- **Same planning rule.** TenneT's IP2026 grid model adds only projects in the realisation phase (IP2026 s5.3); so does ours. After the 4 kA fix our 2035 380/220 kV project set matches TenneT's (RenewaFLEX note G9).
- **Projects not in our grid but due by 2035** (preparation or study phase, 7): Maasbracht-Boxmeer-Dodewaard 380 kV verzwaren verbinding (Voorbereiding, 2031-2034); Geertruidenberg-Rilland 380 kV verzwaren verbinding (Voorbereiding, 2032-2035); Dodewaard-Doetinchem 380 kV verzwaren verbinding (Voorbereiding, 2030-2031); Beverwijk-Vijfhuizen-Bleiswijk 380 kV verzwaren verbinding (Studie, 2030-2031); Eemshaven Oudeschip-Eemshaven 380 kV realiseren 3e circuit (Voorbereiding, 2031-2034); A9-Zuid 380/150 kV realiseren station (Voorbereiding, 2032-2035); Einighausen 380 kV realiseren station (Voorbereiding, 2031-2034).
- **The congested corridors get their new lines after 2035** in TenneT's plan (Geertruidenberg-Krimpen/Crayestein 2037-2040, Diemen-Lelystad and Lelystad-Ens 2038-2041, Vierverlaten-Ens 2035-2038), so leaving them out of a 2035 grid follows TenneT (G6).
- **Ratings:** IP2026 s7.4 says all domestic 380/220 kV connections go to 4 kA within the plan horizon; our 2035 grid applies 4 kA only where a project names the line. Dutch 380 kV corridors still below 4,000 A: Hollandse Kust Zuid Landstation-Maasvlakte (1,170 A), Eemshaven Converterstation-Eemshaven (1,425 A), Bleiswijk-Wateringen (2,470 A), Boxmeer-Dodewaard (2,498 A), Boxmeer-Maasbracht (2,498 A), Dodewaard-Maasbracht (2,500 A), Eemshaven Oudeschip-Vierverlaten (2,500 A), Eindhoven-Tilburg (2,884 A), Geertruidenberg-Tilburg (2,884 A), Dodewaard-Doetinchem (2,910 A), Doetinchem-Hengelo (2,910 A), Hengelo-Zwolle (2,910 A), Beverwijk-Vijfhuizen (3,000 A), Bleiswijk-Vijfhuizen (3,000 A), Borssele-Rilland (3,238 A), Geertruidenberg-Rilland (3,238 A).
- **Method differences** (G9): we reduce to 32 nodes (TenneT: full 380/220/150/110 kV grid); one rating per line (TenneT: seasonal); N-1 transfer limits (TenneT: hour-by-hour n-0/n-1/n-2 checks); 150/110 kV congestion is not visible in our model.

9. Files

What	Where
Workbooks	inputs/derived/jao_core_v29-vvltr2/, inputs/derived/jao_core_v29-vvltr2-2035-tennet-tlb3-rt4k/ (build_report.md, row_decisions.csv, corrections.csv inside)
Patches with sources	inputs/patches/ (CHANGELOG.md, patch_metadata.csv per patch)
Lineage, sources	docs/LINEAGE.md, docs/SOURCES.md, docs/provenance.md
Earlier maps (superseded)	docs/map_nl_grid_v29_v2.pdf, map_nl_grid_v29-2035-*_v2.pdf
RenewaFLEX comparison	nl_merge/G6, G8, G9; nl_merge/figures/grid_vs_tennet_plans_2035.pdf
This report	scripts/map_grid_sources_v3.py -> grid_map_sources_2026-10-26_v3.pdf (+ PNG maps)